CCS6334
DIGITAL & COMPUTER FORENSICS

Lecture 12
Validating the Evidence

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Learning Objectives

After studying this chapter, you should be able to:

- Explain the nature and risks of unsound digital evidence
- Apply principles of impartiality and structured, balanced analysis
- Describe formal evidence validation processes
- Demonstrate effective presentation of validated digital evidence

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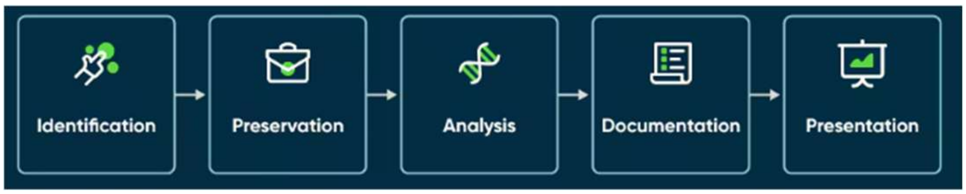
Overview of Forensic & Evidence Validation



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Definition of Forensic Validation

- Forensic validation is the process of confirming that digital forensic tools, methods, and procedures produce accurate, reliable, repeatable, and legally defensible results.
- In simple terms, it answers the question:

👉 *"Can the evidence and findings be trusted in a court of law?"*

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Key Aspects of Forensic Validation

1. **Tool Validation**
   - Ensures forensic tools (e.g., EnCase, FTK, Autopsy, Foremost) function correctly.
   - Confirms that the tool:
     - Extracts data accurately
     - Does not alter original evidence
     - Produces consistent results across repeated runs
2. **Process Validation**
   - Verifies that forensic procedures follow accepted standards.
   - Examples:
     - Proper imaging of storage devices
     - Correct handling of volatile and non-volatile data
     - Maintaining chain of custody

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Key Aspects of Forensic Validation (Cont.)

3. **Result Validation**
   - Confirms that analysis results are correct and reproducible.
   - Often achieved through:
     - Cross-tool verification (using more than one tool)
     - Hash value comparison (MD5, SHA-256)
     - Peer review of findings
4. **Evidence Validation**
   - Ensures evidence remains unchanged from acquisition to presentation.
   - Uses cryptographic hashes to prove integrity.

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Validating Digital Evidence

- Digital forensics lacks a universally accepted validation model, creating weaknesses in practice
- Although *"every case is unique"*, many characteristics of digital evidence are common and reusable
- Core purpose of validation:
  - Test authenticity, relevance, and reliability of evidence
  - Ensure evidence withstands legal scrutiny
- Common pitfalls reduce:
  - Admissibility
  - Evidentiary weight
- Validation requires more than intuition — it demands systematic testing and checking

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Why Evidence Validation Is Important

- Digital evidence is fragile and easily altered
- Courts require proof of evidence integrity
- Unvalidated evidence may be challenged or rejected
- Ensures investigator credibility
- Supports reproducibility of forensic findings
- Objectives of Evidence Validation:
  - Preserve evidence integrity
  - Prevent accidental or intentional modification
  - Confirm correctness of analysis
  - Ensure compliance with forensic standards
  - Support admissibility in court

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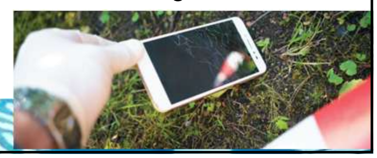
Evidence Contamination

- Digital evidence is highly vulnerable to tampering and undetectable modification, raising serious concerns about authenticity and reliability.
- Unlike physical evidence, altered digital data may leave no visible trace, making original evidence impossible to recover once compromised.
- Courts increasingly recognize these risks, similar to concerns raised over DNA evidence falsification.
- The use of validated forensic tools and proper procedures reduces contamination risks and supports evidentiary reliability.
- Ongoing challenges remain due to the lack of universally accepted standards in digital forensics.
- Proper preservation of digital evidence is critical and must be strictly enforced throughout the investigation lifecycle.



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Impartiality in selecting evidence

- Digital forensic examination must meet the same rigorous standards as other forensic disciplines.
- Practitioners must conduct impartial, unbiased, and thorough investigations guided by a strong ethical code.
- Requires sound forensic experience, continuous learning, and collaboration with other experts.
- Competence in using validated forensic tools and understanding relevant laws and regulations is essential.
- Examiners must recognize their limitations and seek specialist support when required.
- Effective communication with legal teams and courts is critical to clearly justify methods, findings, and evidence validity.

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Maintaining Impartiality and Professional Integrity

- Forensic practitioners must demonstrate complete impartiality and avoid examiner bias or selective evidence interpretation.
- Loss of objectivity leads to reputational damage, turning experts into client-serving witnesses rather than servants of the court.
- Ethical risks include complacency, overconfidence, and indifference, which undermine forensic credibility.
- High-quality digital forensic work demands patience, focus, and professional dedication, even with small datasets.
- Evidence presentation must always be based on sound, methodical examination, regardless of perceived importance.
- Time constraints, budget limits, heavy workloads, and distressing case content increase stress and can threaten forensic quality, requiring strong professional discipline to overcome.

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Maintaining Impartiality – An Example

Scenario:

- A forensic investigator is tasked with examining a company employee's computer to determine if sensitive data was leaked.

What it looks like:

- The investigator examines all files, logs, and evidence objectively without assuming guilt or innocence.
- They do not let personal opinions about the employee, or pressure from management, influence their analysis.
- They follow standard forensic procedures for collecting, preserving, and analyzing evidence.

Example in practice:

- Even if the investigator personally dislikes the employee, they document the findings exactly as they are. If no evidence of wrongdoing is found, the report reflects that, rather than trying to "find something" to support management's suspicions.

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Faulty Case Management & Evidence Validation

- Reexamination of digital evidence has shown that poor analysis and unclear presentation can significantly affect judicial outcomes.
- Two recurring issues: faulty case management and incomplete or incorrect evidence validation.
- Causes include heavy caseloads, inexperience, ignored or unidentified exculpatory evidence, and evidence cherry-picking.
- Digital evidence can be seductive and misleading; practitioners must avoid bias and rely strictly on verifiable facts.
- Inadequate validation risks failed prosecutions or wrongful convictions, undermining justice.
- Robust, formal evidence validation processes are essential to ensure accuracy, relevance, completeness, and temporal consistency, upholding the presumption of innocence.

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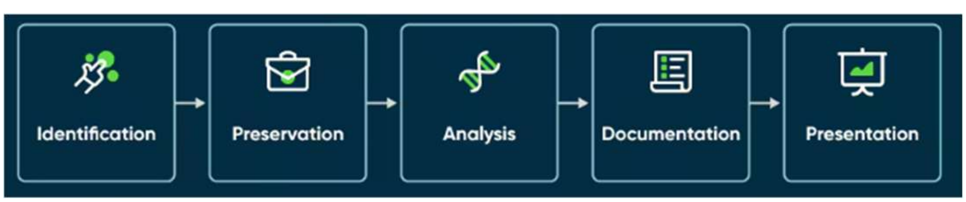
The structured and balanced analysis of digital evidence



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Overview

- In any legal proceedings intuition is not enough and does not impress the court, solid facts are needed and should be supported with logical analysis
- Attempts must be made to locate all evidence, and intuition alone may not be sufficient for an inexperienced practitioner to locate hidden and hard-to-find evidence.
- The incomplete identification of all evidence that should/would/could be located can ruin an examination → possibly due to the incompetence or inexperience of a practitioner or because of the lack of time and available resources
- Linking a suspect to incriminating events, assuming the events are really incriminating, is the first hurdle in any examination, and this requires hypothesis testing

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Developing Hypotheses in Digital Forensics

- Informal internal investigations often lead to evidence contamination and weak forensic outcomes.
- Hypotheses about events are typically formed by investigators or legal teams, not solely by practitioners.
- Forensic practitioners must test, corroborate, or refute each hypothesis using objective digital evidence.
- A professional approach includes exploring alternative and counter-hypotheses to reduce bias.
- Standardized measurement methods and SOPs help improve evidence reliability and reduce human error.
- Examinations must remain evidence-led, objective, and framework-driven, avoiding subjective or guilt-seeking analysis.

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Modeling Arguments

- What properties are necessary/sufficient for evidence to be viable in a specific investigative content?
- The integrity of the data or assuring that the evidence is not modified, intentionally or accidentally, is usually a primary concern
- The ability to authenticate information must be present
- Being able to reproduce the processes used to gather and examine evidence is yet another consideration
- It is essential to know that the seizure of the evidence does not substantively change the state of either the evidence or system from which it was taken.
- It may also be important to demonstrate that only the information relevant to the investigation was accessed

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Toulmin Model of Argumentation

- Proposed by Stephen EdelstonToulmin, a British philosopher, author, and educator who focused his work on the analysis of moral reasoning
- The Toulmin model is a simple representation of the evidence and has been used to show the ultimate hypothesis, allowing each group of evidence to be deconstructed in separate diagrams to show the likelihood of the claim and reservation (or counter-argument)
- The diagram contains six interrelated components used for analyzing arguments, was first published in his 1958 book The Uses of Argument, was considered his most influential work, particularly in the field of rhetoric and communication, and in computer science → it requires the practitioner to show, in visual form, the key issues and counter-issues such that the layman can comprehend

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Toulmin Model of Argumentation

Toulmin's theory defines six aspects common to any type of argument

Toulmin Component	Forensic Application	Example
Claim	Conclusion based on evidence	Suspect accessed confidential files
Data	Collected digital evidence	Server logs, timestamps, IP address
Warrant	Logical reasoning linking data to claim	IP and access times match suspect activity
Backing	Standard principles supporting warrant	Forensic guidelines on log integrity
Qualifier	Certainty of conclusion	“Strongly suggests” or “likely”
Rebuttal	Potential alternative explanations	Someone else used the workstation

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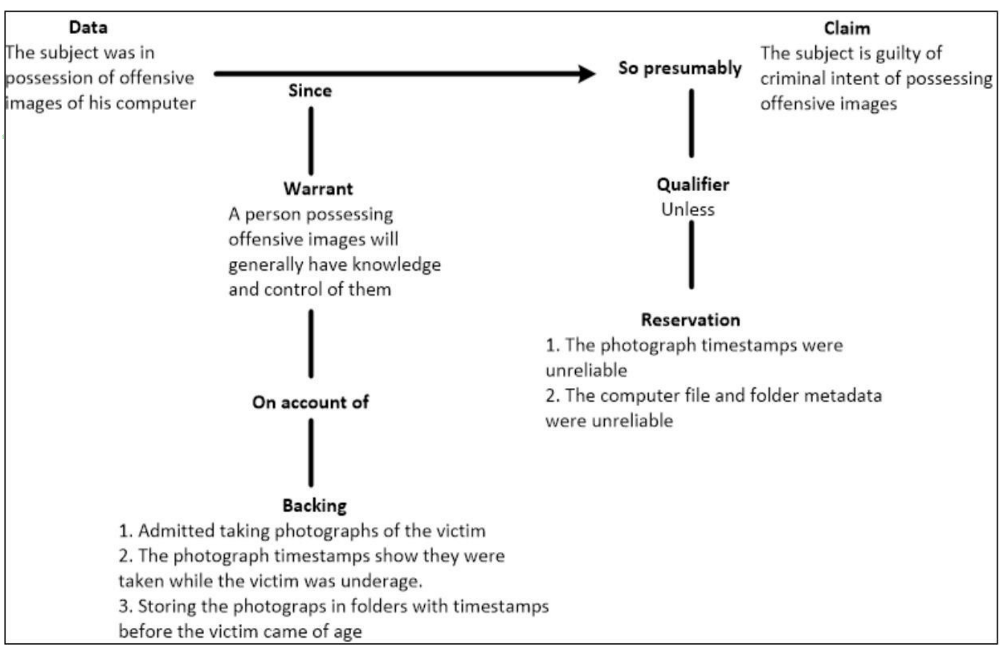

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Bayesian Network Model

- A Bayesian Network is a probabilistic graphical model that represents dependencies among variables. For evidence validation, it helps quantify the likelihood that a piece of evidence is valid, given prior knowledge and other corroborating or conflicting evidence.

Advantages BN in digital evidence validation:

- Handles uncertainty systematically
- Combines multiple evidence sources
- Accounts for reliability and credibility factors
- Supports incremental updates as new evidence is obtained
- Can be used for decision support in investigations

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Rationale for Selecting the Bayesian Network Model

- Selected from multiple reasoning models due to its strength in handling uncertainty in complex digital evidence.
- Widely used to support forensic experts in explaining and validating evidence in legal contexts.
- Acts as a normative model for evaluating new evidence, clearly separating information from opinion.
- Complements abductive reasoning used by courts by providing a scientific, probabilistic foundation for evidence assessment.
- Helps distinguish scientific reasoning (Bayesian probability) from judicial judgment, enhancing evidence interpretation in digital cases.

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Benefits and Application of Bayesian Reasoning

- Provides a structured evidence validation process to identify, test, and measure the reliability of digital exhibits.
- Produces transparent and meaningful validity measurements, improving explainability.
- Serves as an effective training tool for novice forensic practitioners.
- Uses likelihood ratios to express evidentiary weight:
- Probability of evidence given guilt vs. given innocence

$$\frac{\textit{The probability of finding the evidence given that the defendant is guilty}}{\textit{The probability of finding that same evidence given that the defendant is innocent}}$$

- Admissibility vs. inadmissibility of evidence

$$\frac{\textit{The probability of finding that the evidence is admissible}}{\textit{The probability of finding that evidence is inadmissible}}$$

- Combines likelihood ratios with prior knowledge to derive posterior odds, supporting objective assessment of probative value and aiding fair legal decision-making.

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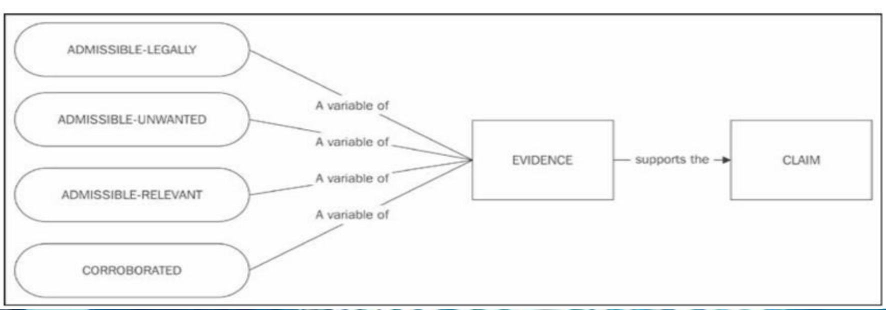
Conceptual Framework of the Proposed Model

- In the model, validity of each evidence object must be checked before it can be considered admissible
- These conditions of the admissibility of digital evidence (and circumstantial evidence in general) may be defined as (1) its legal admissibility, plus its (2) state of being untainted, (3) its relevance to the issues at hand, and, ideally, whether (4) it is corroborated – shown in the following diagram



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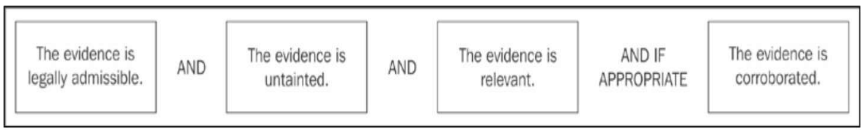
Conceptual Framework of the Proposed Model

- Therefore, for the purpose of the model, all conditions of admissibility must be fulfilled, namely, if the evidence is legally admissible, untainted, and relevant, then the evidence may be admitted and, if appropriate, the evidence should be independently corroborated
- In the following diagram, the highlighted conditions of admissibility and their interdependence are shown



Conditions of admissibility schema

- Further elaboration of the diagram may include details into subsets acceptable in each condition as shown in the figure on the next slide

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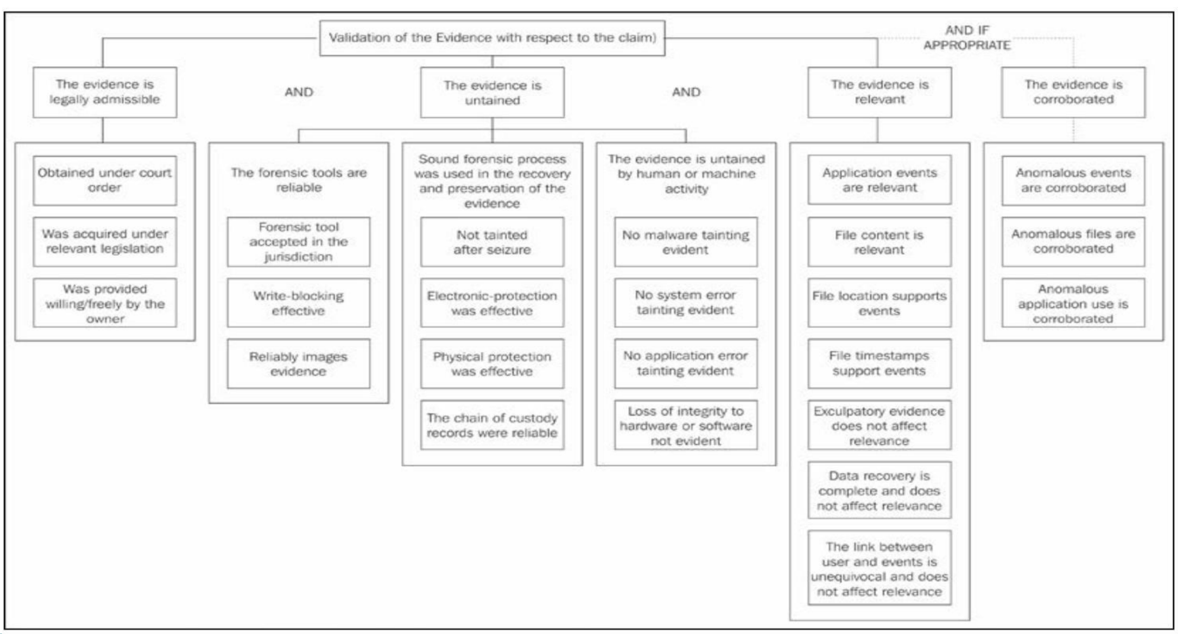

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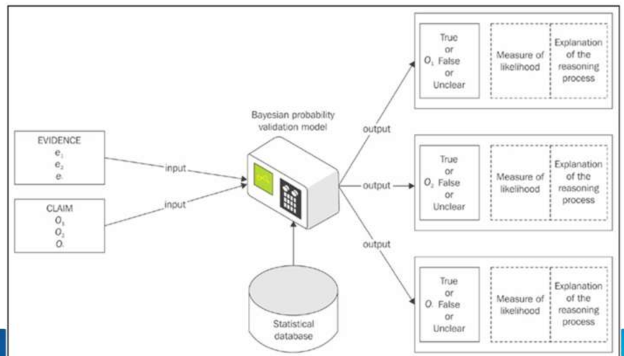
Bayesian Model for Digital Evidence Validation

- Treats digital evidence as structured inputs to a Bayesian reasoning model.
- Evidence is processed through predefined questions and tests linked to a statistical knowledge base.
- Validation involves systematic interrogation against thresholds for relevance and admissibility.
- Supports multiple decision outcomes: Yes, No, Unsure, Discounted, or Cannot Be Determined.
- Produces clear reasoning explanations and recommends further investigation when evidence validity is weak.



The Enhanced Validation Model for Digital Forensic Examinations

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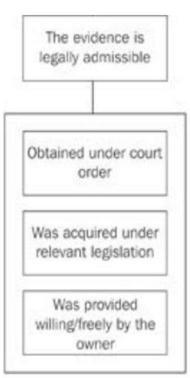
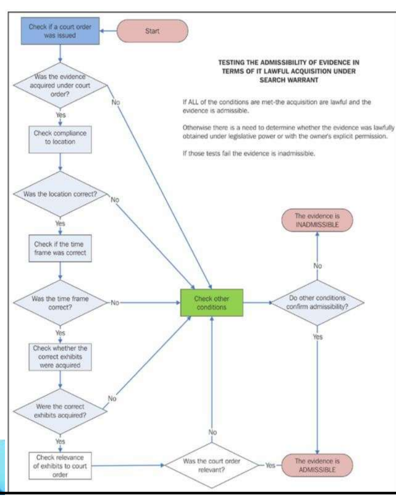
Example of Questions to Validate Conditions

- How can be determine if evidence is legally admissible? From the model, it was shown that if could be obtained under court order, under relevant legislation or provided willingly without the need for previous two conditions





- The process of checking the legitimacy of evidence collected pursuant to a search warrant is set out in the flowchart shown in the following diagram.
- The process requires that all checks are met; otherwise, the information gathered may not be admissible
- The chart on the left shows the possible avenues to take into consideration when contemplating if evidence complies to being obtained under court order

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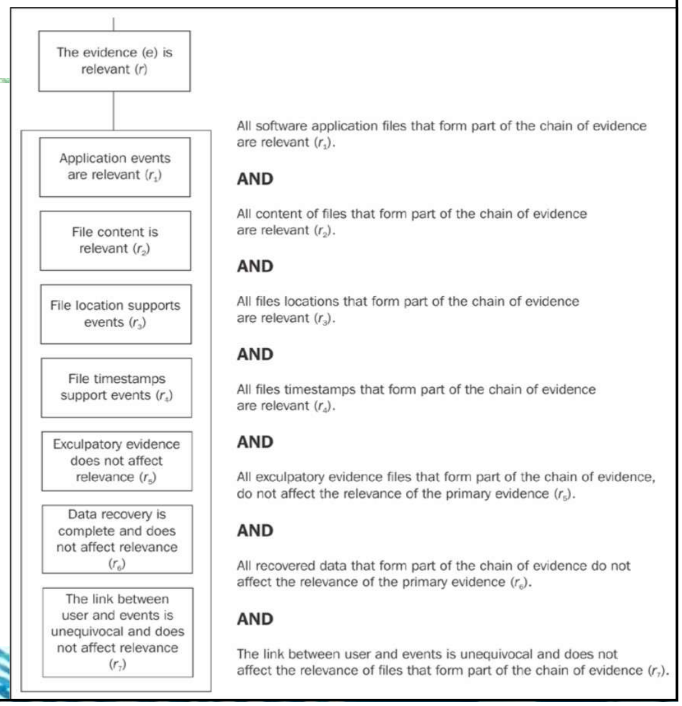
- Some conditions require multiple measurements to verify its admissibility – often all at the same time
- E.g. to prove relevance of evidence often requires all the subconditions listed to be true/valid
- Some subconditions require further conditions to be valid, e.g. R_7 here which states link between users and events is unequivocal (i.e. indisputable) but in some cases it might …..



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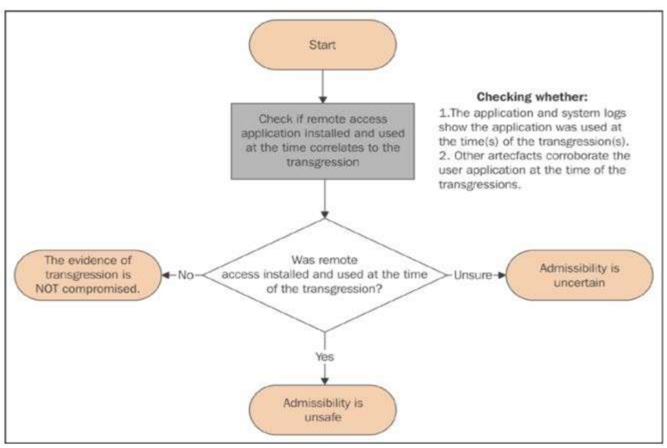

- One possibility discussed before is whether the suspect's device was used to perform a transgression but not under the actions of the suspect/owner, i.e. remotely controlled or manipulated
- Presence of remote software (RDC, VNC, etc) or certain malware could indicate the possibility of this and impact the admissibility of evidence

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Calculating Probabilities – (Pre-Probability)

- Each hypothesis has a prior probability of being true before anything is known about the evidence associated with it
- If the occurrences are infrequent, a low threshold of prior probability (0.01?) is assigned and opposite applies when high prior probability is expected → A lower threshold tells us that such an occurrence is infrequent, whereas a higher threshold suggests greater frequency
- E.g. the probability of someone dying in an airplane crash is lower than being struck by lightning (1/16m vs 1/14k)
- Probability of being infected by malware X if operating system is on version Y, etc.

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Calculating Probabilities – (Post-Probability)

- Post probabilities are ones based on alternate hypotheses that need to be taken into consideration
  - E.g. we have a suspect, their equipment, and trace of transgression – the post probability would measure the likelihood someone framing the suspect of the transgression
- The same measures are applied to indicate likelihood of these alternate hypotheses
  - E.g. probability malware was the cause of the transgression is 1 in 100 (0.01) for insignificant OR 0.9 if trace of a exploit being used is found
- These probabilities can be combined alongside to calculate the overall likelihood of an event occurring
- Example on next slide shows possible hypothesis calculation table in a case contemplating whether the application used was remotely controlled

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Application status and odds instance	The remote application was running during the period of the transgression	Other system data shows that the remote application was running during the period of the transgression	Prior odds (posterior odds 0.8 and 0.01)	Odds observed by the model calculations	Prognosis of remote application compromise
A	No	No	0.1	2 in 443	This hypothesis is discounted
			0.01	1 in 2,427	
B	Yes	No	0.1	9 in 14	This hypothesis is uncertain
			0.01	8 in 57	
C	No	Yes	0.1	9 in 14	This hypothesis is uncertain
			0.01	8 in 57	
D	Yes	Yes	0.1	712 in 713	This hypothesis is possible
			0.01	66 in 67	
E	Uncertain	Uncertain	0.1	1 in 100	This hypothesis is discounted
			0.01		
F	Yes	Uncertain	0.1	53 in 59	This Hypothesis is uncertain
			0.01	17 in 38	
G	Uncertain	Yes	0.1	53 in 59	This hypothesis is uncertain
			0.01	17 in 38	
H	No	Uncertain	0.1	7 in 319	This hypothesis is uncertain
			0.01	1 in 491	
I	Uncertain	No	0.1	7 in 319	This hypothesis is uncertain
			0.01	1 in 491	

- The hypothesis is discounted: Instances A and E show that the hypothesis is highly unlikely.
- Instances B, C, F, G, H, and I should suggest to the evaluator that there is now some uncertainty as to whether a remote exploit had occurred.
- The hypothesis is possible: From these results, the practitioner would recognize that in one instance (D), which reflects that both questions were positive inputs, there was a high likelihood of a remote exploit.
- I.e. imagine as a truth table!!

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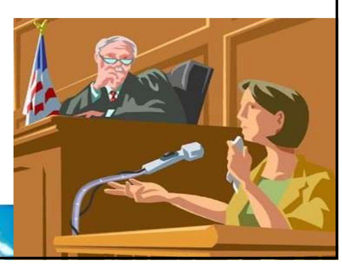
Presentation of Evidence

- There are two main types of testimony given by professionals at a trial, deposition, or hearing:
  - (1) Technical or scientific witness testimony: Provides factual, objective testimony by describing technical procedures, observations, and data obtained using standard tools without offering opinions or interpretations.
  - (2) Expert witness testimony: Offers professional opinions and interpretations based on specialized knowledge and experience to help the court understand the significance and implications of technical evidence.
- It is important to reiterate that in presenting a report or testifying in person, the practitioner must ensure that exculpatory (to clear of guilt) as well as inculpatory (to prove guilt) evidence is presented to the other party. The practitioner must make the following things certain when presenting the evidence:
  - Conclusions are technically sound
  - The evidence solidly supports them and is properly preserved
  - Any exculpatory evidence that may have been found has been considered



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Digital Forensic Reporting

- It is important for forensic practitioners to aim for clarity and simplicity and be certain that the evidence is easy to access and properly cross referenced
- Reports should make recommendations, including those for preventing a recurrence of the incident, and describe the root causes, if any, that permitted it to occur. It is sensible to present the evidence in the best possible light and to demonstrate the examination processes progressively, e.g.
  - The most-favored position first and last with alternatives in the middle
  - Summarizing the evidence in the context of the process
  - Addressing anomalies before the other party does
  - Drawing conclusions, if any, and providing the basis for those conclusions
  - Addressing other possible interpretations and their basis
  - Clear conclusions including understandable and justifiable reasoning
  - Debate whether the results and conclusions change if any of the information given is incorrect or changes

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Digital Forensic Reporting

- A sample forensic report layout is shown here and note it contains an EXECUTIVE SUMMARY meant to outline the contents and purpose of the report, the processes involved, and the analysis of evidence recovered and any other key facts
- The report should also include the evidence-recovery process, including chain-of-custody information if a separate register has not been maintained

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A specimen forensic report layout

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Court Appearances

- Juries place enormous weight on practitioners' evidence, no thanks to CSI
- Just because practitioners know their field does not mean the juries and legal teams will understand
- Cardinal rules for experts are know your stuff, never underestimate lawyers, think before answering, be honest, and know the outcome does not matter
- Practitioners are supposed to be impartial and are expected to justify what scientific principles underpin the processes used, process or validation data for the technique, reference databases used and the reason for using them, and their ability to calculate the statistics and likelihood ratios
- Practitioners should always avoid jargon, attempt to remain within bounds of their expertise, and ensure their research on a specific matter is up to date

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Court Appearances

- Any errors must be reported to legal team immediately and if a practitioner as a change of mind, they must be prepared to change their stance if that is what the truth requires (i.e. admit wrongs)
- Admission to a lack of knowledge or uncertainty and accept other reasonable propositions is a possible path to take and try not to get defensive
- Be wary of multi-part and ambiguous questions that might be misleading
- Legal teams usually will assist in the preparation of expert witness testimony in court as well as the presentation of evidence where necessary

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Ethical Issues Confronting Forensic Practitioners

- Ethics is concerned with right and wrong behavior and makes us reflect on how a person should act in various situations when an ethical dilemma arises
- Some examples of ethical issues – discrimination, harassment, privacy, corporate espionage/sabotage, etc.
- Ethics is actually very hard to define, and different interpretations and environments complicate attempts to define a universal meaning.
- Each of us may derive our own personal ethics from many sources, including family and culture, religion or faith, the legal system, and where we live
- Practitioners might experience situations that test their ethical standards
- Impartiality and service to the court may sound quaint, but it is a solid barricade to defend when pushed to drop standards
- Maintaining a code of ethics allows practitioners to keep their objectivity during the investigation. If you cannot be impartial, you should not be a party to the investigation

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Word Of Advice

- Never assume something to be true without verification – it pays to be informed and up-to-date on information
- Laws that apply in a country may not be the same in other countries
- It is important for practitioners to always be knowledgeable on their domain and not rely on hearsay
- What you see on screen in TV and movies usually do not apply (they have been dramatized for effect)

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The Future Reality of Computer (Digital) Forensics

- Global cybercrime losses exceeded USD 16 billion in 2024, showing a sustained upward trend into 2025 due to large-scale fraud, ransomware, and phishing.
- Phishing and social-engineering attacks remain the dominant attack vector, increasingly powered by AI-generated content, QR-based scams, and deepfakes, significantly increasing success rates.
- Cryptocurrency-related crime continues to rise, with multi-billion-dollar thefts annually, posing major forensic challenges due to pseudonymity, cross-border transactions, and rapid fund laundering.
- Despite rising threats, many organizations especially SMEs still focus security spending on prevention, with limited investment in forensic readiness and post-incident evidence handling.
- Digital evidence is now central to the vast majority of criminal and civil investigations, with a growing proportion originating from mobile devices, cloud platforms, IoT systems, and decentralized environments.
- This reality drives an urgent demand for advanced digital forensics expertise capable of producing evidence that is reliable, objective, accurate, and legally admissible in complex, data-rich environments.

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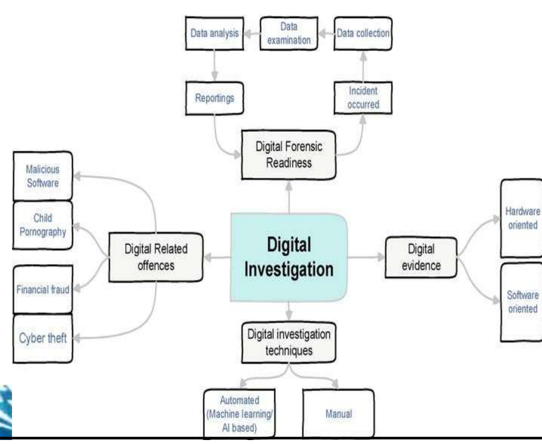
The Future Reality of Computer (Digital) Forensics

- The use of forensics in the foreseeable future would involve a lot of cross-domain expertise
  - Large amounts of digital data may require the expertise of big data experts
  - Handling multiple files (numbering thousands or millions of files) may require the use of automated systems trained using AI/ML methods



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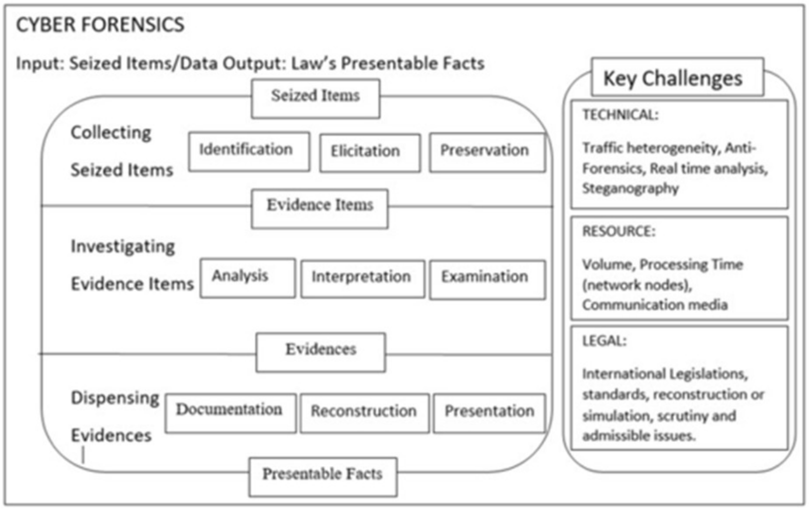
- We are now seeing proposals for forensic frameworks that incorporate these domains together



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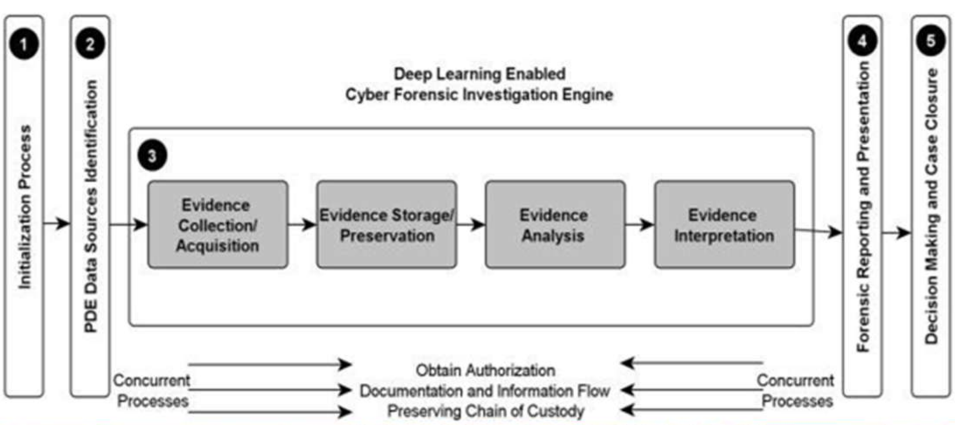
Interception of evidence on mobile networks

- What does this mean for you??
- The need to be constantly updated with latest developments not only on the legal side but also on the tech side
- The need for cross-domain knowledge – or at least know who to ask



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Summary

What was covered today

- The problems of unsound digital evidence, the need for impartiality in undertaking forensic examinations, and some common pitfalls that diminish the value of digital forensics through cursory and biased examinations
- The pressing need for structured and balanced analysis was outlined, as was the need to validate evidence to determine its relevance and authenticity in anticipation of it being tendered in legal proceedings
- A proposed model of validation was presented, intended to benefit practitioners in handling complex evidence and quantifying the results empirically

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THANKS – Q&A

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